

Week 1 Assignment

What is the Role of a Software Quality Assurance (QA) Engineer in the Software Development Life Cycle (SDLC)?

A **Software Quality Assurance (QA) Engineer** is a professional responsible for ensuring that a software application meets the required quality standards before it is released to users. Their primary role is to identify defects, verify that the software meets the specified requirements, and ensure that users receive a reliable, secure, and user-friendly product.

A QA engineer is not only involved during the testing phase. They participate throughout the **Software Development Life Cycle (SDLC)** by working closely with developers, business analysts, designers, project managers, and other stakeholders to ensure quality is built into the software from the beginning of the project's startn.

The roles of a Software QA Engineer at each phase of the SDLC are explained below.

1. Planning Phase

During the planning phase, the QA engineer understands the project's objectives, scope, and expected outcomes. They participate in planning meetings, identify possible risks that may affect software quality, and contribute to developing a testing strategy. At this stage, they also estimate the time, tools, and resources that will be required for testing.

2. Requirements Analysis Phase

In this phase, the QA engineer carefully reviews the software requirements to ensure they are complete, clear, and testable. If any requirement is unclear or incomplete, they communicate with the project team to seek clarification. Understanding the requirements early helps prevent misunderstandings and reduces defects during development.

For example, if the requirement states that users should be able to log into an application, the QA engineer ensures the requirement clearly explains how users will log

in, what should happen if incorrect details are entered, and what error messages should be displayed.

3. Design Phase

During the design phase, the QA engineer reviews the system design to ensure it aligns with the project requirements. They begin preparing test scenarios, writing test cases, and identifying the test data that will be needed. Proper preparation at this stage makes the testing process more organized and effective.

4. Implementation (Development) Phase

As developers begin writing the code, the QA engineer works alongside them by preparing the testing environment and reviewing completed features. They ensure that the software is progressing according to the requirements and make any necessary updates to the test cases if changes are introduced during development.

5. Testing Phase

The testing phase is where the QA engineer performs the actual testing of the software. They execute test cases, identify defects, document bugs, and report them to the development team for correction.

Depending on the project's needs, the QA engineer may perform different types of testing, such as Unit Testing, Integration Testing, System Testing, Acceptance Testing, Smoke Testing, Sanity Testing, Regression Testing, and Re-testing. After developers fix reported issues, the QA engineer verifies that the defects have been resolved without affecting other parts of the application.

6. Deployment and Maintenance Phase

Before the software is released to users, the QA engineer performs final verification to ensure the application is stable and ready for deployment. After deployment, they continue to support the software by testing bug fixes, verifying reported issues from users, and performing regression testing whenever updates or new features are introduced. This helps maintain the quality and reliability of the software throughout its lifecycle.

Conclusion

A Software Quality Assurance Engineer plays an essential role in every stage of the Software Development Life Cycle. Rather than simply finding bugs, they help prevent defects, improve software quality, troubleshoot, and ensure that the final product meets both business requirements and user expectations. By participating from the planning stage through deployment and maintenance, QA engineers contribute significantly to the successful delivery of high-quality software.

WEEK 2 ASSIGNMENT

Question:

Choose any application (web or mobile). Imagine you are the QA Engineer. Identify five non-functional defects that could negatively affect the user experience, even if the application's features work correctly. Add screenshots of the defects found.

Non-Functional Defect Report

Application Tested: OPay Mobile Application

Introduction

Non-functional defects are issues related to the quality of an application, such as usability, accessibility, user interface consistency, readability, and performance. They do not stop the application from functioning, but they can reduce user satisfaction and the overall quality of the product.

1: Missing Space in Greeting

Category: UI / Text Formatting

Observation

On the Home screen, the greeting is displayed as **"Hi,OCHANYA"** instead of **"Hi, OCHANYA."**

Expected Result

The greeting should display with proper spacing after the comma.

Actual Result

The greeting is displayed without a space after the comma.

Impact

Although this does not affect the application's functionality, it reduces the professionalism and visual quality of the user interface.

11:07



Hi,OCHANYA



✓ Available Balance

Transaction History >

₦52.32 >

+ Add Money



Transfer to POS Transfer-MAR... -~~₦~~3,100.00

Jul 22nd, 10:42:21

Successful



Bonus from Airtime Purchase

Jul 22nd, 10:25:48

+~~₦~~4.00

Successful



To OPay



To Bank



Withdraw



Airtime



Data



Betting



TV



)

2: Duplicate "My BizPayment" Menu

Category: UI Consistency

Observation

On the **Me** page, I noticed that **My BizPayment** appears twice with the same icon and description. I clicked on both menu items and confirmed that they both open the same page.

Expected Result

Each menu item should appear only once unless there is a business reason for displaying it multiple times.

Actual Result

The application displays the same menu item twice.

Impact

This duplication may confuse users and create an inconsistent user interface.

11:09



Hi, OCHANYA



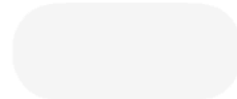
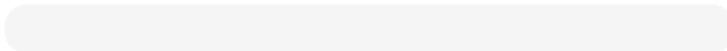
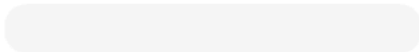
Upgrade to Tier 3



Total Balance 

~~₦~~93.16

Interest Credited Today ~~₦~~+0.96



Transaction History



Account Limits

View your transaction limits



Bank Card/Account

Add payment option



My BizPayment

Receive payment for business



My BizPayment

Receive payment for business



Finding 3: High Notification Count

Category: Usability

Observation

The Home screen displays **45 unread notifications**.

Expected Result

Important notifications should be easy to identify without users feeling overwhelmed.

Actual Result

A high notification count may cause users to ignore notifications, including important account alerts.

Impact

Users may miss important security or transaction updates.

Screenshot: *(Insert Home Screen Screenshot)*

Finding 4: Busy Home Screen

Category: Usability

Observation

The Home screen displays many services and options at the same time.

Expected Result

The interface should allow users to locate frequently used services quickly and easily.

Actual Result

The large number of visible services may make navigation slightly difficult for first-time users.

Impact

Users may spend extra time searching for the feature they need.

Screenshot: *(Insert Home Screen Screenshot)*

11:07



Hi,OCHANYA



✓ Available Balance 

Transaction History >

₦52.32 >

+ Add Money



Transfer to POS Transfer-MAR... -~~₦~~3,100.00

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TV



Finding 5: Small Secondary Text

Category: Accessibility

Observation

Some supporting text under menu items is displayed in a small, light-gray font.

Expected Result

Supporting text should be large enough and have sufficient contrast to be easily readable.

Actual Result

The text may be difficult to read for users with visual impairments or in bright lighting conditions.

Impact

This may reduce readability and accessibility, even though the application's features work correctly.

Screenshot: *(Insert Me Page Screenshot)*

11:09



Hi, OCHANYA



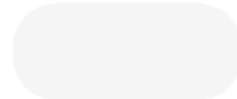
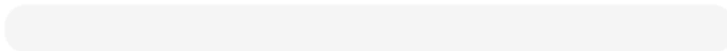
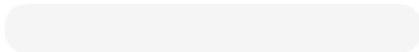
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My BizPayment

Receive payment for business



My BizPayment

Receive payment for business



Conclusion

The OPay mobile application performs its core functions successfully. However, during my review, I identified five non-functional issues related to text formatting, user interface consistency, usability, and accessibility. While these issues do not prevent users from completing transactions or using the application's features, addressing them would improve the overall user experience and make the application more user-friendly.

WEEK 3 ASSIGNMENT

Title: Responsiveness Testing of the NYSC Registration Portal

Objective

The purpose of this assignment is to evaluate the responsiveness of the **NYSC Registration Portal** on different screen sizes. Responsiveness testing helps determine whether a website automatically adjusts its layout, text, images, and navigation to provide a good user experience on different devices.

For this assignment, I tested the website using:

- **Desktop (Laptop View)**
 - **Mobile Device (iPhone 13 Pro)**
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What is Responsiveness Testing?

Responsiveness testing is a type of testing that checks whether a website displays and functions properly on different screen sizes and devices.

A responsive website should automatically adjust its layout without affecting usability, readability, or functionality.

As a QA Engineer, responsiveness testing helps ensure that users have a consistent experience whether they visit the website on a laptop, tablet, or mobile phone.

Application Tested

Website: NYSC Registration Portal

Devices Used

- Desktop (Laptop Browser)
 - Mobile Device (iPhone 13 Pro)
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Test Observation

1. Desktop View

Observation

The website displayed very well on the desktop screen.

Findings

- All service cards were displayed side by side.
- The text was clear and easy to read.
- Buttons such as **Verify Certificate**, **Payment Status**, **Certificate Replacement**, **Senate List Check**, **Biometric Registration**, and **NYSC Main Website** were visible at the same time.
- Navigation was easy because many options could be accessed without scrolling.
- Images and icons appeared sharp.
- The layout looked neat and well organized.

Result: Responsive and user-friendly.

2. Mobile View (iPhone 13 Pro)

Observation

The website also adapted to the smaller screen size.

Findings

- The service cards were arranged vertically instead of horizontally.
- I had to scroll down to access each service.
- All buttons were still functional.
- The text remained readable, although it appeared smaller than on the desktop.
- I had to pay more attention to read some of the text because of the smaller screen size.
- The navigation menu collapsed into a hamburger menu, which saved screen space.
- No overlapping text or broken layout was observed.

Result: Responsive and functional.

Comparison Between Desktop and Mobile

Desktop	Mobile
Multiple services displayed at once	Services displayed one after another
Larger text	Smaller text
Easy to navigate without much scrolling	Requires more scrolling
Wider layout	Narrow vertical layout
Very comfortable to read	Readable but slightly more difficult because of screen size

Overall Result

The NYSC Registration Portal is responsive across all devices tested.

Although the mobile version requires more scrolling and the text is smaller due to the limited screen size, the website maintains its functionality and usability.

The layout adjusts properly, navigation remains accessible, and no responsiveness defects such as overlapping elements, broken buttons, or distorted images were observed during testing.

Conclusion

Based on my testing, I conclude that the NYSC Registration Portal provides a good responsive design on both desktop and mobile devices.

The website successfully adapts to different screen sizes while maintaining its functionality and ease of use. The only noticeable difference is that mobile users need to scroll more and may find the smaller text slightly less comfortable to read compared to the desktop version.

WEEK 4 ASSIGNMENT

Boundary Value Testing (BVT)

Task 1: Identify the Boundaries

Rule 1: Wallet Funding (Per Transaction)

Requirement:

- Minimum funding amount: **₦1,000**
- Maximum funding amount: **₦200,000**

Scenario	Test Value	Expected Result
Below Minimum	₦ 999	Reject. Display: <i>"Minimum funding amount is ₦1,000."</i>
At Minimum	₦ 1,000	Funding should be successful.
Above Minimum	₦ 1,001	Funding should be successful.
Below Maximum	₦ 199,999	Funding should be successful.
At Maximum	₦ 200,000	Funding should be successful.
Above Maximum	₦ 200,001	Reject. Display: <i>"Maximum funding amount per transaction is ₦200,000."</i>

Rule 2: Daily Funding Limit

Requirement:

- Maximum total funding per day: **₦500,000**

Scenario	Daily Total	Expected Result
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Below Maximum	₦499,999	Funding should be allowed.
At Maximum	₦500,000	Funding should be allowed if this transaction brings the total exactly to ₦500,000.
Above Maximum	₦500,001	Reject. Display: <i>"Daily funding limit of ₦500,000 exceeded."</i>

Rule 3: BVN Field

Requirement:

- Must contain **exactly 11 digits**
- Only numeric characters are allowed

Scenario	Test Value	Expected Result
Below Minimum Length	1234567890 (10 digits)	Reject. Display: <i>"BVN must be exactly 11 digits."</i>
At Required Length	12345678901 (11 digits)	Accept if all characters are numeric.
Above Maximum Length	123456789012 (12 digits)	Reject. Display: <i>"BVN must be exactly 11 digits."</i>
Contains Letters	12345AB8901	Reject. Display: <i>"BVN must contain only numeric characters."</i>
Contains Special Characters	12345@78901	Reject. Display: <i>"BVN must contain only numeric characters."</i>

Task 2 & 3: Critical Thinking Scenarios

1. A user funds ₦200,000 three times in one day. What should happen on the third attempt?

Answer:

- First transaction: ₦200,000 Accepted
- Second transaction: ₦200,000 Accepted
- Total so far: ₦400,000
- Third transaction: ₦200,000 → Total would become ₦600,000

Expected Result:

The third transaction should be rejected because it exceeds the daily funding limit of ₦500,000.

2. A user funds ₦499,000 in total, then attempts ₦2,000 more. What should happen?

Current total = ₦499,000

New transaction = ₦2,000

Total = ₦501,000

Expected Result:

The transaction should be rejected because the total exceeds the daily funding limit.

3. A user enters BVN: 1234567890

This BVN contains **10 digits**.

Expected Result: Reject the input.

Error Message:

BVN must be exactly 11 digits.

4. A user enters BVN: 12345678901

This BVN contains **11 numeric digits**.

Expected Result:

Accept the BVN because it meets the validation requirements.

5. Where is the system most likely to fail per transaction limit or cumulative daily limit? Why?

The system is **more likely to fail at the cumulative daily limit**.

This is because the system must keep track of all successful funding transactions throughout the day and continuously calculate the total amount funded. If the application does not update or validate the running total correctly, it may incorrectly allow users to exceed the ~~N~~500,000 daily limit

Week 5 Assignment

7th August 2026

: Visit this [website](#) and come up with detailed 5 test case IDs, module names, and test scenarios

Test case ID	Module	Test case name	Test scenario
Tc-001	Login	Correct login details	Enter the standard_user and correct password and verify successful login in.

Tc-002	Login	Wrong username	Enter a wrong username with the correct password and check if login is rejected
Tc-003	login	Wrong password	Enter the correct username with a wrong password and check if login is rejected
Tc-004	login	Username left blank	Leave the username empty and check if the system shows an error
Tc-005	login	Password left blank	Leave the password empty and check if the system shows an error